

**Multiple Choice:**

1. Distance near the skip distance should be used for sky-wave propagation
 - A. to avoid tilting
 - B. to prevent sky wave and upper ray interference
 - C. to avoid the Faraday effect
 - D. so as not to exceed the critical frequency
2. Diffraction of electromagnetic wave
 - A. is caused by reflections from the ground
 - B. arises only with the spherical wavefronts
 - C. will occur when the waves pass through a large slot
 - D. may occur around the edge of a sharp obstacle
3. In wave propagation, determine the refractive index of an ionospheric layer with 100×10^6 free electrons per cubic meter at 5 MHz frequency.
 - A. 0.88
 - B. 0.99
 - C. 0.77
 - D. 0.66
4. A range of microwave frequency more easily passed by the atmosphere than the other is called a
 - A. window
 - B. critical frequency
 - C. gyro frequency range
 - D. Resonance in the atmosphere
5. What causes the ionosphere to form?
 - A. Lightning ionizing the outer atmosphere
 - B. Release of fluorocarbons into the atmosphere
 - C. Temperature changes ionizing the outer atmosphere
 - D. Solar radiation ionizing the outer atmosphere
6. In skywave propagation, if the critical frequency is 25 MHz, what is the best frequency to use? Assume radiation angle of 45° .
 - A. 30.00 MHz
 - B. 25.25 MHz
 - C. 35.35 MHz
 - D. 20.20MHz
7. A wave that is characterized by having its direction of propagation perpendicular to its oscillation is known as
 - A. Isotropic
 - B. Transverse
 - C. Polarized
 - D. Refractive
8. Determine the critical frequency value of an HF signal if its maximum usable frequency is 7050.50 kHz at 35 degrees incidence.
 - A. 5775 kHz
 - B. 5577 kHz
 - C. 5557 kHz
 - D. 7755 kHz
9. High frequency waves are
 - A. absorbed by the F2 layer
 - B. reflected by the D layer
 - C. capable of use for long distance communication on the moon
 - D. affected by the solar cycle
10. The position of the E layer in the ionosphere is:
 - A. above the F layer
 - B. below the F layer
 - C. below the D layer
 - D. sporadic
11. Skip distance is a term associated with signals from the ionosphere. Skip effects are due to:
 - A. selective fading of local signals
 - B. high gain antennas being used
 - C. local cloud cover
 - D. reflection and refraction from the ionosphere
12. What power density is required to produce electric field strength of 100 volts per meter in air?
 - A. 25.6 W/m²
 - B. 26.5 W/m²
 - C. 22.8 W/m²
 - D. 27.3 W/m²
13. Which is not an effect of our environment on wave propagation?
 - A. Radiation
 - B. Reflection
 - C. Refraction
 - D. Diffraction
14. How are VHF signals propagated within the range of the visible horizon?
 - A. By plane wave
 - B. By geometric wave
 - C. By direct wave
 - D. By sky wave
15. Electromagnetic waves are refracted when they
 - A. pass into a medium of different dielectric constant
 - B. are polarized at right angles to the direction of propagation
 - C. encounter a perfectly conducting surface
 - D. pass through small slot in a conducting plane
16. Which ionospheric region is closest to the Earth?
 - A. The D region
 - B. The E region
 - C. The F region
 - D. The A region
17. An unknown medium was used in communication system. If the characteristic impedance of that medium is 150 ohms, what is the relative permittivity?
 - A. 1.51
 - B. 3.51
 - C. 2.51
 - D. 4.51
18. How are VHF signals propagated within the range of the visible horizon?
 - A. By plane wave
 - B. By geometric wave
 - C. By direct wave
 - D. By sky wave



19. Indicate which one of the following terms applies to troposcatter propagation:

- A. SIDs
- B. Fading
- C. Atmospheric storms
- D. Faraday Rotation

20. A mobile radio system base station is installed 50 meters above the ground and transmits 150 W at 54 MHz. If the transmitting antenna gain is 5.17, how far will the receiving antenna be, if the antenna height is 30 m and the received field strength is 22 $\mu\text{V/m}$.

- A. 151.313 km
- B. 155.213 km
- C. 153.313 km
- D. 150.213 km

21. The ground wave eventually disappears, as one moves away from the transmitter, because of

- A. interference from the sky wave
- B. loss of line of sight conditions
- C. maximum single hop distance limitations
- D. tilting

22. The wavelength of a radio signal is:

- A. equal to f/c
- B. equal to $c \times f$
- C. the distance a wave travels in one period
- D. how far the signal can travel without distortion

23. How far away is the radio horizon of an antenna 100 ft high?

- A. 14.14 mi
- B. 41.41 mi
- C. 24.24 mi
- D. 10.10 mi

24. A point in space that radiates electromagnetic energy equally in all directions is called

- A. Transverse
- B. Isotropic point source
- C. Omnisphere
- D. Shadow zone

25. How are VHF signals propagated within the range of the visible horizon?

- A. By plane wave
- B. By geometric wave
- C. By direct wave
- D. By sky wave

END